Chapter 18: Comparing Two Means

The Practice of Statistics in the Life Sciences
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Part IV: Specific Inference Procedures

Why Chapter 18 Matters

Chapter 17 developed inference for one population mean and for paired differences.

Many scientific questions instead compare two separate populations or two independent treatment groups.

Main Question

How can two independent samples be used to estimate or test the difference between two population means?

The main parameter is now a difference, $\mu_1 - \mu_2$, rather than a single mean.

Chapter Roadmap

The chapter develops the following connected ideas.

1. Recognize independent two-sample problems
2. Describe two groups before performing inference
3. Derive the standard error of $\bar{x}_1 - \bar{x}_2$
4. Construct Welch two-sample t intervals and tests
5. Assess robustness to non-Normality
6. Understand why pooled procedures and inference about standard deviations require caution

The reasoning is the same as in earlier chapters: study design first, then conditions, calculation and interpretation.

What Is a Two-Sample Problem?

A two-sample problem compares a quantitative response across two distinct groups.

- There is one sample from each population or treatment group
- The same quantitative response is measured in both groups
- The individuals in one sample are unrelated to those in the other sample
- The two sample sizes may be different

Goal

Estimate or test the difference between the population means, $\mu_1 - \mu_2$.

How Two Independent Samples Arise

Two-sample data can come from several designs.

- **Randomized experiment:** Experimental units are randomly assigned to two treatments
- **Separate random samples:** One random sample is selected from each of two populations
- **Subgroups of one sample:** A random sample is divided using a pre-existing characteristic such as species, habitat or sex

The design determines whether the comparison can support a causal conclusion, a population generalization, both or neither.

Independent Samples or Matched Pairs?

The correct procedure depends on how the observations were produced.

	Independent samples	**Matched pairs**
Relationship	Different, unrelated individuals	Same individual twice or deliberately matched units
Analysis	Compare $\bar{x}_1 - \bar{x}_2$	Calculate paired differences first
Procedure	Two-sample t	One-sample t on differences
Example	Two separate chicken groups	Before-and-after blood pressure

Ignoring a real pairing discards information. Inventing a pairing between unrelated observations is also incorrect.

What Conclusions Does the Design Support?

Design feature	**What it supports**	**What it does not guarantee**
Random sampling	Generalization to the sampled population	Causation
Random assignment	Causal comparison of treatments	Generalization beyond the experimental units
Both	Causal inference and population generalization	Protection from noncompliance or measurement problems
Neither	Association in the observed data	Broad generalization or causation

A small P-value cannot compensate for a weak study design.

A Quantitative Response and a Two-Level Factor

A two-sample problem can be viewed as a bivariate problem.

- The **response variable** is quantitative
- The **explanatory variable** is categorical with two levels

Example

In a study of a feed supplement and calf growth, weight gain is the quantitative response and supplement status (supplement or control) is the two-level factor.

The statistical question concerns the difference between the two population mean responses.

Examples of Two-Sample Questions

Study	Response	Two groups
Triclosan exposure in mice	Liver weight	Triclosan diet; control diet
Young chimpanzee behaviour	Time in contact with mother	Male; female
Pertussis records	Time since vaccination	Diagnosed; not diagnosed
Transgenic chickens	Hatching weight	Transgenic; commercial

In every case, the same quantitative variable is compared across two independent groups.

Describe the Data Before Inference

Begin with side-by-side graphs and numerical summaries.

- Use dotplots for small samples
- Use boxplots or histograms for larger samples
- Compare shapes, centres and spreads
- Look for skewness, clusters, gaps and outliers
- Check whether a mean is scientifically meaningful for each group

Why This Comes First

The two-sample t procedure summarizes a difference in means. A graph can reveal features that the two means cannot describe.

Conditions for Comparing Two Means

Welch two-sample t procedures rely on three main ideas.

1. Each sample represents its population or treatment group through a defensible random process
2. The two samples are independent of one another
3. Each population is Normal, or the samples are large enough for the procedure to be reliable without strong outliers

The two population means and standard deviations are unknown. Equal population variances are **not** required.

Independence Has Two Levels

The design should support independence both within and between samples.

- **Within a sample:** One observation should not determine another
- **Between samples:** An individual in Group 1 should not be paired or otherwise linked with an individual in Group 2
- When sampling without replacement, each population should be much larger than its sample

Common Violation

Measurements taken before and after treatment on the same animals are paired, not independent. They require a matched pairs analysis.

Shape, Outliers and Sample Size

The reliability of the procedure depends on both groups.

- With small samples, each distribution should be reasonably symmetric and free of outliers
- Moderate samples can tolerate mild skewness
- Larger samples can tolerate more skewness because of the Central Limit Theorem
- Extreme outliers can strongly affect a sample mean and standard deviation even when the other group is well behaved

Assess the two distributions separately. One acceptable graph does not repair a serious problem in the other group.

Population Notation

Each group has its own population mean and standard deviation.

Population	Variable	Mean	Standard deviation
1	X_1	μ_1	σ_1
2	X_2	μ_2	σ_2

Parameter of Interest

The population difference is $\mu_1 - \mu_2$. Its sign depends on which group is labelled Group 1.

Define the order before calculating or interpreting the difference.

Sample Notation

The two samples provide estimates of the four unknown population quantities.

Sample	Size	Mean	Standard deviation
1	n_1	$\bar{x}_1$	s_1
2	n_2	$\bar{x}_2$	s_2

The natural estimate of $\mu_1 - \mu_2$ is

$$\bar{x}_1 - \bar{x}_2$$

This estimate is unbiased when the samples are produced by appropriate random processes.

The Difference Must Keep Its Order

Suppose Group 1 is the treatment group and Group 2 is the control group.

$$\text{estimated treatment effect} = \bar{x}_1 - \bar{x}_2$$

- A positive value means the treatment sample mean is larger
- A negative value means the treatment sample mean is smaller
- Reversing the group labels reverses the sign but not the strength of the evidence

State the subtraction order in words before interpreting an interval or test statistic.

Centre of the Sampling Distribution

Across repeated independent samples,

$$E(\bar{X}_1) = \mu_1 \qquad \text{and} \qquad E(\bar{X}_2) = \mu_2$$

Therefore,

$$E(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2$$

Interpretation

The difference in sample means is an unbiased estimator of the difference in population means.

Why the Variances Add

For independent random variables, the variance of a difference is the sum of the variances.

$$\text{Var}(\bar{X}_1 - \bar{X}_2) = \text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2)$$

Thus,

$$\text{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

The variances add even though the means are subtracted. Uncertainty from both independent samples contributes to the difference.

Standard Error of a Difference in Means

The population standard deviation of the sampling distribution is

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Because σ_1 and σ_2 are unknown, estimate them with s_1 and s_2:

$$SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Larger sample sizes reduce the standard error. Greater within-group variation increases it.

The Welch Two-Sample t Statistic

To standardize an observed difference, compare it with a hypothesized population difference Δ_0.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

For the common null hypothesis of equal means,

$$H_0 : \mu_1 - \mu_2 = 0 \qquad \Rightarrow \qquad \Delta_0 = 0$$

The statistic measures how many estimated standard errors the observed difference is from the null difference.

Welch–Satterthwaite Degrees of Freedom

The reference distribution is approximately a t distribution with

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{s_2^2}{n_2}\right)^2}$$

- The degrees of freedom need not be an integer
- Software uses the unrounded value
- For hand calculations with a table, rounding down is conservative

What the Degrees of Freedom Reflect

Welch's degrees of freedom depend on both sample sizes and both sample standard deviations.

$$\min(n_1 - 1, n_2 - 1) \leq df \leq n_1 + n_2 - 2$$

- Similar group variances and balanced sample sizes tend to give larger df
- Unequal variances and unequal sample sizes can reduce df
- Larger df makes the t curve closer to the Standard Normal curve

The formula accounts for the uncertainty introduced by estimating two population standard deviations.

Two-Sample t Confidence Interval

A level C confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where t^* comes from the Welch t distribution with the calculated degrees of freedom.

Interpretation

We are C% confident that the population mean for Group 1 is between the reported lower and upper amounts above the population mean for Group 2.

If the entire interval is negative, describe how much lower Group 1 is instead.

Two-Sample t Test

The most common null hypothesis is

$$H_0 : \mu_1 - \mu_2 = 0$$

The test statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

The P-value is an area under the Welch t distribution. A small P-value indicates that a difference at least this extreme would be unusual if the population means were equal.

The Alternative Determines the Tail

Alternative	***P*-value area**
$H_a : \mu_1 - \mu_2 > 0$	Right of the observed t
$H_a : \mu_1 - \mu_2 < 0$	Left of the observed t
$H_a : \mu_1 - \mu_2 \neq 0$	Both tails beyond $\lvert t \rvert$

Choose the alternative from the research question before examining the sample difference.

A Four-Step Two-Sample Analysis

1. **State:** Define the two populations and the parameter $\mu_1 - \mu_2$
2. **Plan:** Select a Welch two-sample t interval or test and define the subtraction order
3. **Solve:** Graph both samples, check conditions, then calculate the standard error, degrees of freedom and result
4. **Conclude:** Interpret the evidence or interval in context and respect the study design

Reminder

Statistical significance, effect size and causal interpretation are different questions.

Example 18.2: Daily Activity and Body Type

Researchers monitored 20 healthy adults who did not regularly exercise. Ten were lean and ten were mildly obese.
Sensors recorded the daily number of minutes spent standing or walking.

Question

Do lean adults spend more time standing or walking, on average, than mildly obese adults?

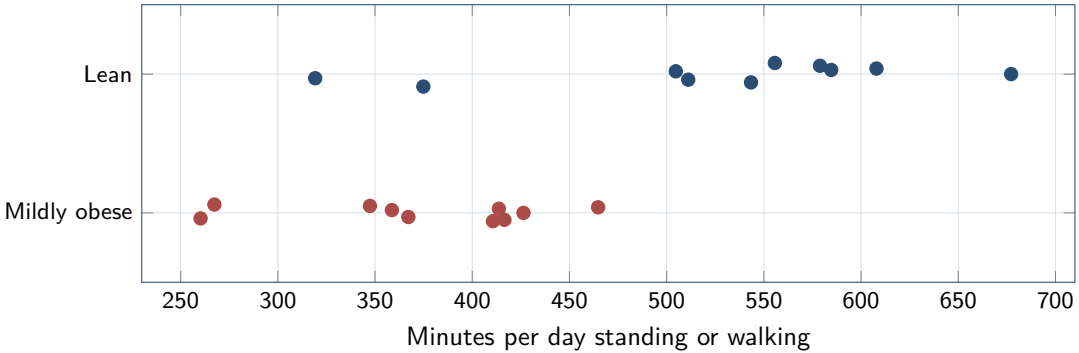
Let Group 1 be lean adults and Group 2 be mildly obese adults. The parameter is $\mu_1 - \mu_2$.

Example 18.2: Observed Data

Lean: minutes/day	Mildly obese: minutes/day
511.100	260.244
607.925	464.756
319.212	367.138
584.644	413.667
578.869	347.375
543.388	416.531
677.188	358.650
555.656	267.344
374.831	410.631
504.700	426.356

The samples are small, so the individual observations should be examined before inference.

Example 18.2: Compare the Distributions



The lean sample is shifted to the right. Neither group shows an extreme outlier or severe skewness.

Example 18.2: Conditions and Scope

- The participants were volunteers rather than simple random samples
- Researchers used eligibility criteria to make the two groups reasonably comparable
- The two groups contain different individuals and can be treated as independent
- The dotplots show no severe skewness or extreme outliers

Scope

The calculation can compare similar lean and mildly obese adults, but generalization to all adults requires caution. The observational design does not establish causation.

Example 18.2: Hypotheses

The researchers expected the lean group to spend more time standing or walking.

$$H_0 : \mu_{\text{lean}} - \mu_{\text{obese}} = 0$$

$$H_a : \mu_{\text{lean}} - \mu_{\text{obese}} > 0$$

This is a right-tailed Welch two-sample t test. The direction must be specified before using the observed difference.

Example 18.2: Sample Summaries

Group	n	$\bar{x}$	s
Lean	10	525.751	107.121
Mildly obese	10	373.269	67.498

The observed difference is

$$\bar{x}_1 - \bar{x}_2 = 525.751 - 373.269 = 152.482 \text{ minutes/day}$$

The lean participants averaged about 152.5 additional minutes of standing or walking per day in this sample.

Example 18.2: Standard Error

$$SE = \sqrt{\frac{107.121^2}{10} + \frac{67.498^2}{10}}$$

$$= \sqrt{1147.491 + 455.598} = \sqrt{1603.089} \approx 40.039$$

The estimated standard deviation of the difference in sample means is about 40.0 minutes per day. Both groups contribute to this uncertainty.

Example 18.2: Degrees of Freedom

$$df = \frac{(1147.491 + 455.598)^2}{\frac{1147.491^2}{9} + \frac{455.598^2}{9}} \approx 15.17$$

Software uses $df \approx 15.17$. A printed table could use $df = 15$.
The value is below $n_1 + n_2 - 2 = 18$ because the sample standard deviations differ.

Example 18.2: Test Statistic

Under H_0, the hypothesized difference is 0.

$$t = \frac{152.482 - 0}{40.039} \approx 3.808$$

The observed difference is about 3.81 estimated standard errors above the null difference.
Because the alternative is right-tailed, the P-value is the area to the right of $t = 3.808$ under a t distribution with approximately 15.17 degrees of freedom.

Example 18.2: P-Value

$$P(T_{15.17} \geq 3.808) \approx 0.00084$$

If the two population means were equal, a sample difference at least this large in the predicted direction would occur in about 0.084% of comparable repeated studies.

Decision at $\alpha = 0.05$

Because $0.00084 < 0.05$, reject H_0.

The calculation provides strong evidence of a positive difference in population means under the model.

Example 18.2: Conclusion

The data provide very strong evidence that the population mean daily standing or walking time is greater for lean adults than for mildly obese adults similar to those studied.

This conclusion should be separated into three statements.

- **Evidence:** The P-value is very small
- **Direction:** The lean population mean is estimated to be larger
- **Limitation:** The observational volunteer study cannot determine whether body type causes activity differences

Example 18.2: Welch Test in R

R code

```
lean <- c(511.100, 607.925, 319.212, 584.644, 578.869,
          543.388, 677.188, 555.656, 374.831, 504.700)
obese <- c(260.244, 464.756, 367.138, 413.667, 347.375,
           416.531, 358.650, 267.344, 410.631, 426.356)
t.test(lean, obese,
       alternative = "greater",
       var.equal = FALSE)
```

The default R procedure is Welch's test. Writing var.equal = FALSE makes that choice explicit.

Why Add a Confidence Interval?

The test answers whether the evidence against equal means is strong. It does not directly show how large the difference may be.

Estimation Question

How much greater is the population mean daily standing or walking time for lean adults?

A confidence interval reports a range of plausible values for $\mu_{\text{lean}} - \mu_{\text{obese}}$ and is more informative about practical importance.

Example 18.4: Confidence Interval Calculation

For $df \approx 15.17$, the 95% critical value is approximately

$$t^* \approx 2.129$$

The margin of error is

$$m = 2.129(40.039) \approx 85.255$$

Therefore,

$$152.482 \pm 85.255 = (67.227,\ 237.737)$$

Example 18.4: Confidence Interval in R

R code

```
activity_result <- t.test(
        lean, obese,
        alternative = "two.sided",
        conf.level = 0.95,
        var.equal = FALSE)
activity_result[["estimate"]]
activity_result[["conf.int"]]
```

For a two-sided 95% interval, use alternative = "two.sided". The interval estimates the difference in the order supplied to t.test().

Example 18.4: Interpret the Interval

We are 95% confident that lean adults similar to those studied spend, on average, between approximately 67 and 238 more minutes per day standing or walking than mildly obese adults similar to those studied.

- The interval contains only positive values
- This agrees with rejection in a two-sided test at $\alpha = 0.05$
- The interval is wide because the samples are small and individual activity varies substantially

Whether the estimated range is clinically important requires subject-matter judgment.

Example 18.4: Association Is Not Causation

The study found a statistically significant association between body type and activity.

- Participants were not randomly assigned to be lean or mildly obese
- Metabolism, health, occupation, environment or mood could influence both body type and activity
- Reverse direction is also possible: activity may influence body type

The data alone cannot determine the causal direction or rule out confounding variables.

Example 18.5: Transgenic Chickens

Researchers compared hatching weights of two independently selected groups of chickens of the same breed.

- Group 1: 45 transgenic chickens resistant to avian influenza
- Group 2: 54 commercial chickens
- Response: hatching weight in grams

Question

Does the genetic modification appear to change the population mean hatching weight in either direction?

Example 18.5: Conditions and Hypotheses

Side-by-side graphs show no strong skewness or clear outliers. The two groups contain different chickens and can be treated as independent.
There was no directional expectation before examining the data, so use a two-sided alternative.

$$H_0 : \mu_{\text{transgenic}} - \mu_{\text{commercial}} = 0$$

$$H_a : \mu_{\text{transgenic}} - \mu_{\text{commercial}} \neq 0$$

Example 18.5: Sample Summaries

Group	n	$\bar{x}$ (g)	s (g)
Transgenic	45	45.14	3.32
Commercial	54	44.99	4.57

The observed difference is

$$45.14 - 44.99 = 0.15 \text{ g}$$

The sample means are close, but inference must compare this difference with its standard error.

Example 18.5: Standardize the Difference

Using the displayed rounded summaries,

$$SE = \sqrt{\frac{3.32^2}{45} + \frac{4.57^2}{54}} \approx 0.795 \text{ g}$$

$$df \approx 95.3$$

$$t = \frac{0.15}{0.795} \approx 0.19$$

The observed difference is only about 0.19 standard errors from zero.

Example 18.5: P-Value and Conclusion

For a two-sided alternative,

$$P \approx 2P(T_{95.3} \geq 0.19) \approx 0.85$$

The precise value from the unrounded data is approximately 0.847.

Conclusion

The data do not provide statistically significant evidence that the two population mean hatching weights differ.

This conclusion is inconclusive about exact equality. It is not evidence that the two means must be identical.

Example 18.5: Calculations from Summaries in R

R code

```
n1 <- 45; xbar1 <- 45.14; s1 <- 3.32
n2 <- 54; xbar2 <- 44.99; s2 <- 4.57

se <- sqrt(s1^2 / n1 + s2^2 / n2)
df <- (s1^2 / n1 + s2^2 / n2)^2 /
      ((s1^2 / n1)^2 / (n1 - 1) +
        (s2^2 / n2)^2 / (n2 - 1))

t_stat <- (xbar1 - xbar2) / se
p_value <- 2 * pt(-abs(t_stat), df = df)
c(SE = se, df = df, t = t_stat, P_value = p_value)
```

Example 18.5: Confidence Interval

Using the unrounded data, the 95% confidence interval for $\mu_{\text{transgenic}} - \mu_{\text{commercial}}$ is approximately

$$(-1.424,\ 1.731)\ \text{g}$$

- A modest negative difference is plausible
- Zero difference is plausible
- A modest positive difference is plausible

The interval explains why the test cannot identify a direction for the population difference.

A Large P-Value Does Not Prove Equality

A large P-value means the observed data are compatible with the null model. It does not assign a high probability to H_0.

- The study may have limited precision
- Several population differences can remain compatible with the data
- An equivalence claim requires a scientifically meaningful tolerance and a procedure designed to test equivalence

Preferred Language

Fail to reject H_0; do not say that the test proves the means are equal.

What Does Robust Mean?

A procedure is **robust** when its confidence level or P-value changes little under moderate violations of its distributional conditions.

Welch two-sample t procedures are often robust to moderate non-Normality, especially with larger samples.

Important Limitation

Robustness to shape does not correct dependence, selection bias, measurement bias or confounding.

Practical Sample-Size Guidance

Assess each group separately, using the smaller sample size as a cautious guide.

- **Small samples:** Both groups should be reasonably symmetric, single-peaked and free of outliers
- **Moderate samples:** Mild skewness is usually acceptable, but strong skewness or outliers remain concerning
- **Large samples:** Stronger skewness can often be tolerated, although influential outliers still deserve investigation

These are practical guidelines, not automatic rules.

Balance Matters

Two-sample procedures behave best when both sample sizes are reasonably large.

- A large sample in one group does not fully compensate for a very small sample in the other
- The smaller group can dominate uncertainty when it also has greater variation
- Balanced designs often provide better precision for a fixed total sample size

Welch's method remains appropriate when sample sizes and variances differ, but the resulting standard error and degrees of freedom reflect that imbalance.

Outliers Require Investigation

Sample means and standard deviations are not resistant.

- Verify whether an unusual value is a data-entry or measurement error
- If valid, report it rather than removing it merely to obtain significance
- Compare results with and without the value as a sensitivity analysis when appropriate
- Consider a transformation or a method suited to the scientific distribution

Any exclusion rule should be scientifically justified and transparently reported.

A Large Sample Does Not Make Every Mean Useful

With very large samples, a t calculation may be numerically reliable even for a non-Normal distribution.

However, a single mean may still be scientifically misleading.

Example

A bimodal distribution of bee cocoon widths may combine smaller male cocoons and larger female cocoons. One overall mean mixes two biologically distinct groups.

The issue is not only whether the procedure works mathematically. The parameter itself must answer a meaningful scientific question.

Welch and Pooled Procedures Are Different

Two versions of the two-sample t procedure appear in software and older textbooks.

	Welch procedure	**Pooled procedure**
Variance assumption	Allows $\sigma_1^2 \neq \sigma_2^2$	Requires $\sigma_1^2 = \sigma_2^2$
Standard error	Uses s_1 and s_2 separately	Uses one pooled variance estimate
Recommendation	Use by default	Avoid unless equality is well justified

The Pooled Two-Sample Standard Error

The pooled procedure estimates a common population variance using

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

It then uses

$$SE_{\text{pooled}} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

with $df = n_1 + n_2 - 2$. These formulas are valid only under the additional equal-variance assumption.

Why the Pooled Procedure Should Usually Be Avoided

- Equal population variances are rarely known in advance
- Tests of equal variances add another assumption-sensitive decision
- Pooled inference can be inaccurate when variances and sample sizes differ
- Welch's procedure is reliable whether the population variances are equal or unequal
- With equal variances, Welch usually loses little precision

Recommendation

Use Welch's two-sample t procedure as the routine method.

R Uses Welch's Procedure by Default

R code

```
# Recommended: Welch two-sample t procedure
t.test(response ~ group, data = study_data)
# The same choice written explicitly
t.test(response ~ group, data = study_data,
       var.equal = FALSE)
# Pooled procedure: requires equal population variances
t.test(response ~ group, data = study_data,
       var.equal = TRUE)
```

In R, do not change var.equal to TRUE merely because the sample standard deviations look similar.

Avoid Routine Inference about Standard Deviations

The t procedures compare means. They do not test whether two population standard deviations are equal.
Classical procedures for variances often rely much more strongly on Normal population distributions than t procedures do.

- Skewness and outliers can distort variance-based tests substantially
- A few extreme observations can dominate a sample standard deviation
- Statistical significance in spread may not answer the scientific question of interest

Begin by graphing the distributions and describing spread in context.

Why Variance Tests Are Fragile

For Normal populations, ratios of sample variances can be compared using an F distribution.

Unlike Welch's mean procedure, the classical F test for equal variances is highly sensitive to non-Normality.

Practical Approach

If variability is the primary scientific outcome, plan the analysis with a method appropriate to the distribution and study design rather than using a routine equal-variance test as a preliminary step.

The decision to use Welch's procedure does not require an equal-variance test.

Choose the Procedure from the Design

Data structure	**Procedure**
One quantitative sample	One-sample t
Same subjects measured twice	Matched pairs t on differences
Two unrelated quantitative samples	Welch two-sample t
Two independent categorical outcomes	A procedure for proportions, not means

The data structure determines the model. The formula should not be chosen only from the number of columns in a data file.

Chapter 18 Summary

- The parameter is $\mu_1 - \mu_2$, estimated by $\bar{x}_1 - \bar{x}_2$
- Independent variances add, giving

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- Welch's degrees of freedom account for two estimated variances
- Tests assess evidence; intervals estimate the size and direction of a difference
- Robustness improves with sample size, but outliers and design problems remain important
- Welch's procedure is preferred to the pooled procedure

Two-Sample t Procedures in R

R code

```
# Explore the two groups first
boxplot(response ~ group, data = study_data)
# Two-sided Welch test and 95% confidence interval
t.test(response ~ group, data = study_data)
# One-sided alternatives
t.test(response ~ group, data = study_data,
       alternative = "greater")
t.test(response ~ group, data = study_data,
       alternative = "less")
```

Check the order of the factor levels because it determines the sign of the reported difference.

Final Analysis Checklist

Before reporting a two-sample comparison, ask:

1. Are the samples truly independent rather than paired?
2. Is the same quantitative response measured in both groups?
3. Do graphs reveal skewness, outliers or important subgroups?
4. Is the subtraction order defined and maintained?
5. Is Welch's procedure being used?
6. Does the conclusion distinguish evidence, effect size, generalization and causation?

Main Principle

The value of a two-sample analysis comes from combining a sound design, an appropriate model and a contextual interpretation.